POWER PULSE

Circuit Breakers Important Formulae

Quick Revision Concepts by BALAJI SIR

1 Ratings

1. Breaking MVA $= \sqrt{3}\, V_L(\text{kV})\, I_b(\text{kA})$ (symmetrical rms at rated voltage)
2. Making current $\approx 2.55 \times I_{b,sym}$ (peak), $\quad I_{sc} = \dfrac{MVA_{base}}{\sqrt{3}\, V\, X_{pu}}$
3. Rated sequence: $O - 0.3\text{ s} - CO - 3\text{ min} - CO$

2 Transient recovery voltage (TRV)

4. $$\omega_n = \frac{1}{\sqrt{LC}}, \quad f_n = \frac{1}{2\pi\sqrt{LC}}$$
5. $$v(t) = E_m\,(1 - \cos\omega_n t), \qquad v_{peak} = 2E_m$$

   E_m = peak phase voltage (undamped)
6. $$RRRV_{max} = E_m\,\omega_n \;\; (\text{at } \omega_n t = \tfrac{\pi}{2})$$
7. $$R_{critical} = \frac{1}{2}\sqrt{\frac{L}{C}}$$ (resistance switching: no oscillation for $R \le R_c$)

   parallel resistor across contacts

3 Overvoltages

8. $$\tfrac{1}{2}LI^2 = \tfrac{1}{2}CV^2 \;\Rightarrow\; V = I\sqrt{L/C}$$

   current chopping (inductive switching)
9. Capacitive switching: restrike can give up to $3\times$ the peak voltage

4 Types

10. ACB, oil (bulk/minimum), air-blast, vacuum, SF_6 (best for EHV)
11. Arc extinction at current zero: high-pressure cooling, high resistance, high-dielectric gas